

Lavish Laller

Backend & Systems Engineer | Java, Spring Boot, Rust

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Education

B.Tech, Computer Engineering

J.C. Bose UST, YMCA Faridabad, India

2022 – 2026

CGPA: 8.21 / 10

Class XII

Govt. Sr. Sec. School, Ladwa (KKR), India

2022

95.4%

Work Experience

Samsung R&D Institute (SRI-D)

Noida, India

Software Engineering Intern

Jan 2026 – July 2026

- Built an on-device AI daemon in **Rust** with an autonomous, fault-tolerant self-diagnosis pipeline for edge system recovery.
- Automated resolution of **40%** of observed system failures through deterministic recovery workflows.
- Cut p95 initialization latency from **800 ms to 480 ms** by benchmarking and tuning Microsoft **Semantic Kernel** under high-load edge workloads.
- Engineered integrations for **MCP (Model Context Protocol)** servers, enabling zero-install execution of third-party services within the native UX.
- Developed a concurrent execution engine mapping natural language to terminal commands over **gRPC** and **Unix Domain Sockets**.
- Tech Stack:** Rust, LLMs, Semantic Kernel, MCP, gRPC, Unix Domain Sockets, Tizen OS

Projects

mcp-wall / OS-Level Sandbox Runtime for Untrusted MCP Servers

[GitHub](#)

- Kernel-enforced sandbox for untrusted MCP servers using Linux **Landlock** and **seccomp**, blocking the filesystem-escape class of **CVE-2025-53109/53110**.
- 2.4 MB** static binary, **3.6 MB** RAM, $\sim 8 \mu s$ per tool call, **0.5 ms** startup; adds a TOCTOU-safe egress filter and DLP redaction of AWS keys and PII.
- Deterministic adversarial suite of **14 attack scenarios** in CI.
- Tech Stack:** Rust, MCP, Linux Kernel (Landlock, seccomp), JSON-RPC, CI/CD

PaperMind / Multi-Tenant RAG Document Q&A Backend

[GitHub](#)

- Multi-tenant RAG service on Spring AI + pgvector (HNSW, cosine) answering from user documents with page-level citations; per-user isolation enforced as a userId filter on every search and asserted by test.
- 2.0–2.5s end-to-end answers streamed over SSE with citations in the first frame; 7/8 graded correct across 3 chunking configs on a reproducible retrieval harness.
- Async ingestion off the request thread; SHA-256 dedupe, type allowlist and 2000-chunk cap reject documents before embedding spend.
- Tech Stack:** Java 21, Spring Boot, Spring AI, PostgreSQL, pgvector, JWT, Docker, REST, SSE

AlgoForge / Multi-Agent AI System for Human-like Problem Solving

[GitHub](#)

- Multi-agent orchestration engine on Microsoft AutoGen with Gemini and OpenAI, decomposing DSA problems across specialized agent roles.
- 3× faster end-to-end solve time than a single-agent baseline via parallel role execution.
- Async FastAPI backend for concurrent task execution and real-time state, exposed as a REST API with Firebase auth.
- Tech Stack:** Python, FastAPI, AutoGen, Gemini API, OpenAI SDK, Firebase, Docker

Technical Skills

Languages: Java, Rust, Python, SQL

Java & Backend: Core Java, OOP, Collections, Exception Handling, Concurrency, Spring Boot, Spring AI, JPA/Hibernate, REST APIs, HTTP/HTTPS, Java HttpClient, RestTemplate, SSE, JWT Auth, PostgreSQL

AI & Developer Tools: GitHub Copilot, Cursor, Prompt Engineering, LLM Agents, Multi-Agent Systems, RAG, Vector Search (pgvector, HNSW), Embeddings, Model Context Protocol (MCP), Semantic Kernel, AutoGen

Networking & Systems: HTTP/HTTPS, REST, URL Structures, TCP/IP, gRPC, JSON-RPC, Unix Domain Sockets, Linux Kernel Internals (Landlock, seccomp), Process Sandboxing, OS Security

Engineering: Data Structures & Algorithms, Operating Systems, System Design, Unit Testing, Debugging, Refactoring, Docker, CI/CD, Git

Honors & Achievements

- Presented on-device AI daemon work to senior leadership visiting from **Samsung headquarters** at an internal AI workshop.
- Contributed data and analysis to a company **town hall** on upcoming technologies and project roadmaps, working with a cross-functional team.
- Solved **500+** problems across LeetCode, Codeforces, and CodeChef.